

Genomics Data: Structure, Formats, and Public Databases

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Outline

Introduction

From Sequencing to Analysis-Ready Data

The Count Matrix

The SummarizedExperiment Object

Public Databases

Quality Assessment

The Afternoon Lab

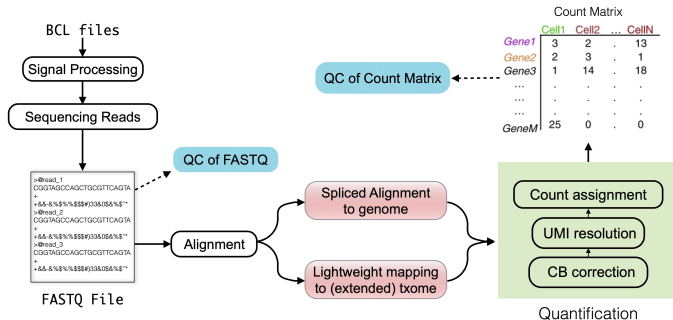
What We Are Doing This Week

- ▶ Over the next three days we move from modeling a handful of variables on a few thousand participants to analyzing **tens of thousands of variables** simultaneously.
- ▶ The biology generates the data. We bring the statistical tools to make sense of it.
- ▶ Today's goal: understand the **structure** of genomics data well enough to work with it in R.
- ▶ We are not bioinformaticians. We will not run alignment pipelines or call variants. We will work with the **analysis-ready data** that those pipelines produce.
- ▶ The statistical challenges — many variables, small sample sizes, correlated measurements, multiple testing — are squarely in our domain.

The Core Statistical Challenge

- ▶ In a typical RNA-sequencing study:
 - ▶ **Variables (genes):** $p \approx 20,000$
 - ▶ **Observations (samples):** $n \approx 6\text{--}50$
- ▶ We are firmly in the regime $p \gg n$.
- ▶ Standard regression and ANOVA methods were designed for $n \gg p$. They do not directly apply.
- ▶ We need methods that:
 - ▶ Test each gene separately (then correct for multiple testing — Day 13)
 - ▶ Account for the count nature of the data
 - ▶ Handle the specific variance structure of sequencing counts
- ▶ The statistical framework for this (Days 12–13) is more tractable than it sounds — but only once we understand the data structure.

The Genomics Data Pipeline



- **We work at the top right.** The earlier steps require specialized bioinformatics software and are typically done before the data reaches us.
- Our job starts with the count matrix.

File Formats: A Brief Orientation

FASTQ

- ▶ Raw sequencing output
- ▶ Contains DNA sequences and quality scores per read
- ▶ Very large (gigabytes per sample)
- ▶ Not human-readable in practice

BAM

- ▶ Aligned reads (binary format)
- ▶ Reads mapped to positions on the reference genome
- ▶ Requires `samtools` to inspect
- ▶ You will rarely touch these

VCF

- ▶ Variant Call Format
- ▶ Lists genetic variants (SNPs, insertions, deletions) vs. the reference genome
- ▶ Relevant for GWAS (not this course)

Count Matrix

- ▶ Genes $\times$ samples table
- ▶ Integer counts per gene per sample
- ▶ This is where we start
- ▶ Familiar structure: rows are variables, columns are observations

What a Count Matrix Looks Like

countData

gene	ctrl_1	ctrl_2	exp_1	exp_1
geneA	10	11	56	45
geneB	0	0	128	54
geneC	42	41	59	41
geneD	103	122	1	23
geneE	10	23	14	56
geneF	0	1	2	0
...	...	...	...	...
...	...	...	...	...
...	...	...	...	...

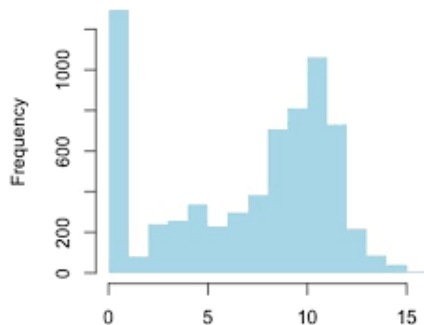
- ▶ Rows: genes (20,000–60,000 in a human dataset)
- ▶ Columns: samples (often 4–50 in a typical study)
- ▶ Values: non-negative integers — how many sequencing reads mapped to that gene in that sample.

Statistical Properties of Count Data

- ▶ Counts are **non-negative integers** — Gaussian models are inappropriate.
- ▶ The distribution is **right-skewed**: a small number of highly expressed genes have very large counts; most genes have low counts; many genes have counts of zero.
- ▶ Variance is **not constant**: highly expressed genes tend to have higher variance (overdispersion).
- ▶ This is why standard linear models and t-tests, applied directly to raw counts, give incorrect results.

The Distribution of Counts

- ▶ The count distribution is well-described by a **negative binomial distribution** — a generalization of the Poisson that allows the variance to exceed the mean.
- ▶ For a Poisson distribution:
 $\text{Var}(Y) = E(Y)$.
- ▶ For a negative binomial:
 $\text{Var}(Y) = \mu + \alpha\mu^2$ where $\alpha > 0$ is the **dispersion parameter**.
- ▶ Estimating the dispersion parameter per gene is one of the main statistical challenges in RNA-seq analysis (covered on Day 12).



The Sequencing Depth Problem

- ▶ Each sample is sequenced to a certain **depth** (total number of reads).
- ▶ If sample A has 20 million reads and sample B has 40 million reads, we expect B's counts to be roughly twice as large — even if the underlying biology is identical.
- ▶ Raw counts are therefore **not directly comparable** across samples.
- ▶ This is a fundamental pre-processing challenge: we must **normalize** for sequencing depth before any comparison.
- ▶ DESeq2 (Day 12) handles this through **size factor estimation** — a single number per sample that accounts for its total sequencing depth.

Sequencing Depth: A Concrete Example



- ▶ Even within the same experiment, total reads per sample can vary substantially.
- ▶ Failing to account for this would confound biological differences with technical ones.

Organizing Genomics Data in R

- ▶ A genomics dataset has three linked components:
 1. **Count matrix:** genes $\times$ samples integer counts.
 2. **Sample metadata:** one row per sample — treatment group, patient ID, sex, age, batch, etc.
 3. **Gene metadata:** one row per gene — gene name, chromosome, genomic coordinates, etc.
- ▶ These three components must stay **synchronized**: if you reorder samples, the sample metadata must reorder too.
- ▶ The Bioconductor SummarizedExperiment object is designed specifically to keep these three things linked and prevent them from getting out of sync.

The SummarizedExperiment Structure

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...	...	...	...	...
...	...	...	...	...
...	...	...	...	...

colData

id	treatment	sex
ctrl_1	control	male
ctrl_2	control	female
exp_1	treatment	male
exp_2	treatment	female

Sample names:

ctrl_1, ctrl_2, exp_1, exp_2

countData is the count matrix
(number of reads mapping to each gene for each sample)

colData describes metadata about the *columns* of countData

First column of colData must match column names of countData (-1st)

Accessing the Components in R

- ▶ Three accessor functions, one for each component:
 - ▶ `assay(se)` — the count matrix
 - ▶ `colData(se)` — sample metadata
 - ▶ `rowData(se)` — gene metadata
- ▶ The object also stores experiment-level metadata (who ran it, what protocol, etc.) in `metadata(se)`.
- ▶ In the afternoon lab, you will spend time with each of these functions on the `airway` dataset.

```
> dim(airway)
[1] 63677      8
> colData(airway)
DataFrame with 8 rows and 9 columns
  SampleName cell dex albut Run avgLength Experiment Sample BioSample
  <factor> <factor> <factor> <factor> <factor> <integer> <factor> <factor> <factor>
SRR1039508 GSM1275862 N61311 untrt untrt SRR1039508 126 SRX384345 SRS508568 SAMN02422669
SRR1039509 GSM1275863 N61311 trt untrt SRR1039509 126 SRX384346 SRS508567 SAMN02422675
SRR1039512 GSM1275866 N052611 untrt untrt SRR1039512 126 SRX384349 SRS508571 SAMN02422678
SRR1039513 GSM1275867 N052611 trt untrt SRR1039513 87 SRX384350 SRS508572 SAMN02422670
SRR1039516 GSM1275870 N080611 untrt untrt SRR1039516 120 SRX384353 SRS508575 SAMN02422682
SRR1039517 GSM1275871 N080611 trt untrt SRR1039517 126 SRX384354 SRS508576 SAMN02422673
SRR1039520 GSM1275874 N061011 untrt untrt SRR1039520 101 SRX384357 SRS508579 SAMN02422683
SRR1039521 GSM1275875 N061011 trt untrt SRR1039521 98 SRX384358 SRS508580 SAMN02422677
> head(assay(airway))
      SRR1039508 SRR1039509 SRR1039512 SRR1039513 SRR1039516 SRR1039517 SRR1039520 SRR1039521
ENS000000000003      679      448      873      408      1138      1047      770      572
ENS000000000005      0      0      0      0      0      0      0      0
ENS000000000019      467      515      621      365      587      799      417      508
ENS000000000047      260      211      263      164      245      331      233      229
ENS0000000000460      60      55      40      35      78      63      76      60
ENS000000000938      0      0      2      0      1      0      0      0
```

Why This Matters Statistically

- ▶ The `colData` is your **study design matrix** — the grouping variable for differential expression, covariates to adjust for, batch indicators.
- ▶ If the `colData` is wrong or misaligned, every downstream analysis is wrong.
- ▶ A common and catastrophic error: the sample order in the count matrix does not match the sample order in the metadata. The `SummarizedExperiment` structure prevents this by design.
- ▶ **Always inspect** `colData` **before modeling**. Know which column encodes treatment vs. control, and verify that it matches what you expect.

Where Genomics Data Lives

- ▶ Most published genomics data must be deposited in a public repository at the time of publication.
- ▶ Three major repositories:
 - ▶ **GEO (Gene Expression Omnibus)** — NCBI's repository for expression data (microarray and RNA-seq). This is the one we use.
 - ▶ **dbGaP** — controlled-access repository for data with individual-level genetic information (requires application and IRB approval).
 - ▶ **SRA (Sequence Read Archive)** — raw sequencing reads (FASTQ/BAM). Requires significant processing before use.
- ▶ For this course: GEO is the right starting point. It provides processed expression matrices and metadata that are immediately usable in R.

GEO: Structure and Vocabulary

- ▶ Each study in GEO has a unique **accession number**:
 - ▶ **GSE** (Series): a complete study (e.g., GSE52778 — the airway experiment).
 - ▶ **GSM** (Sample): a single sample within a study.
 - ▶ **GPL** (Platform): the sequencing or microarray platform used.
- ▶ To use a published dataset: find the GSE number in the paper's data availability section, then download with GEOquery in R.
- ▶ The GEO website (<https://www.ncbi.nlm.nih.gov/geo>) lets you search and browse datasets before downloading.

Data Provenance: Read the Metadata

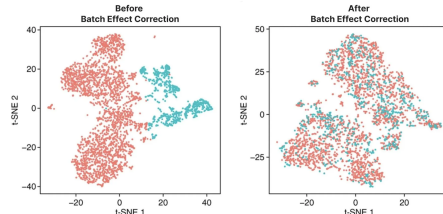
- ▶ **Data provenance** = the documented history of how data were collected, processed, and stored.
- ▶ Before analyzing any public dataset, answer these questions from the metadata:
 - ▶ What biological question was the original study designed to answer?
 - ▶ What organism, tissue, and cell type?
 - ▶ What was the treatment and control condition?
 - ▶ What preprocessing was already applied (alignment, filtering, normalization)?
 - ▶ Are there batch effects? Were multiple sequencing runs combined?
- ▶ This is not a formality. Misreading the metadata is one of the most common sources of error in re-analysis of public data.

Batch Effects: A Statistical Warning

- ▶ A **batch effect** is systematic technical variation introduced by when or how samples were processed — not biology.
- ▶ Common sources: different sequencing runs, different lab technicians, different reagent lots, different dates.
- ▶ If batches are confounded with treatment groups (e.g., all controls run in one batch), batch effects are **impossible to separate** from biological signal.

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- ▶ Always check: is the batch variable in colData? Is it balanced across treatment groups?
- ▶ Batch effects are **a design problem first, a statistical problem second.**

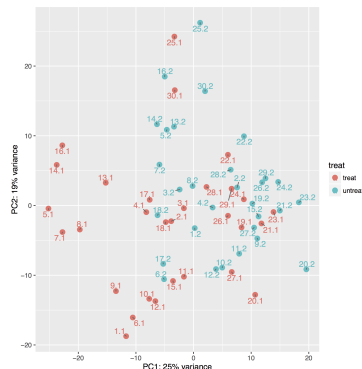


Quality Assessment Before Analysis

- ▶ Just as we run diagnostic plots after fitting a regression model, we run quality checks *before* fitting a genomics model.
- ▶ Key checks:
 1. **Library sizes:** are total read counts per sample roughly comparable?
 2. **Zero-count genes:** how many genes have zero counts across all or most samples? These carry no information and can be filtered.
 3. **Sample-to-sample distances:** do samples cluster as expected by treatment group, or do technical factors dominate?
 4. **Outlier samples:** is any single sample very different from all others in the same group?
- ▶ These are the genomics equivalent of the Day 6 residual diagnostic plots: standard first steps before any modeling.

Checking Sample Similarity: PCA

- ▶ **Principal Component Analysis (PCA)** reduces the 20,000-dimensional count matrix to 2 dimensions that capture the most variation.
- ▶ A well-behaved dataset: samples from the same treatment group cluster together; different groups are separated.
- ▶ Warning signs:
 - ▶ Samples from the same group scatter widely.
 - ▶ Separation is driven by a technical variable (run date, batch) rather than treatment.
 - ▶ One sample is a clear outlier from its group.
- ▶ PCA of genomics data is covered in depth on Day 14. Today: use it as a *diagnostic tool*.



What We Will Do This Afternoon

- ▶ Download a real RNA-seq dataset from GEO using the `GEOquery` package.
- ▶ Inspect the three components of the `SummarizedExperiment`: counts, sample metadata, gene metadata.
- ▶ Verify that the study design (`colData`) matches what the paper describes.
- ▶ Compute basic quality metrics: library sizes, zero-count genes, log-transformed count distributions.
- ▶ Produce a PCA plot and assess whether samples cluster as expected.
- ▶ This is an entirely new kind of data for most of you. The R commands are new. The logic — inspect, summarize, check for problems — is exactly what you have been doing all week with Framingham.

Key Takeaways

1. Genomics data is high-dimensional ($p \gg n$) — this creates statistical challenges not present in standard regression.
2. The analysis-ready object is a count matrix: genes $\times$ samples, integer values. The **upstream bioinformatics pipeline** produces this; we work with what it produces.
3. Counts require **normalization** for sequencing depth and modeling with **negative binomial** or similar count distributions.
4. The SummarizedExperiment keeps counts, sample metadata, and gene metadata synchronized. Always inspect colData before modeling.
5. **Quality assessment first**, modeling second. Check library sizes, zero counts, and sample clustering before fitting any differential expression model.
6. Batch effects are a design problem. If batches are confounded with treatment, no statistical method can fix it.